

Classify different types of energy as kinetic or potential and investigate energy transfer and transformations in simple systems

Learning intention

- classify different types of energy as kinetic or potential and investigate energy transfer and transformations in simple systems.

Teach and model

- observing a Rube Goldberg machine and identifying the energy transfers and transformations involved.
- critiquing and using representations such as flow diagrams to illustrate changes between different forms of energy in a system.
- investigating relationships between kinetic and potential energy in a simple system such as a roller-coaster or Newton's cradle.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

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Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.